# Statistical Arbitrage in Binary Prediction Markets: Three Systematic Strategies and Their Empirical Properties

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**ABSTRACT**

We document three systematic trading strategies deployable on binary prediction markets. **SPREAD_FADE** exploits the structural gap between bid-ask mid-prices and quoted ask prices in near-certain and near-zero probability markets, generating a theoretically riskless edge when the market maker spread exceeds twice the taker fee. **CROSS_MARKET** identifies groups of mutually exclusive outcome markets (tournament brackets, award nomination sets) whose YES prices collectively exceed unity, signalling oversubscription; we short the overpriced outcomes via their complementary NO contracts. **DEP_GRAPH** enforces stochastic dominance constraints on strike-price ladders for crypto assets: a lower strike must price at least as high as any higher strike, and violations constitute pure arbitrage. An ancillary latency sub-strategy fires when a reference asset moves faster than the prediction market reprices. Across 50-cycle synthetic simulations, fractional Kelly sizing produces trade-size distributions consistent with theoretical optima. We derive closed-form edge conditions, analyse failure modes, and discuss deployment constraints including geo-blocked data feeds and minimum depth requirements.

***Keywords:*** *prediction markets, binary options, arbitrage, market microstructure, Kelly criterion, stochastic dominance, spread trading, information efficiency*

**JEL Classification:** G12, G14, G17, D84

*This paper is a working draft circulated for discussion purposes. The strategies described operate on paper trading accounts only. Nothing herein constitutes investment advice. Comments welcome.*

# 1. Introduction

Binary prediction markets occupy a distinctive niche in the landscape of financial markets. Unlike continuous-price securities, each contract resolves to exactly $1 (YES wins) or $0 (NO wins), imposing hard no-arbitrage constraints that are routinely violated in practice. These violations arise from (i) the structural cost of market-making, (ii) the computational difficulty of enforcing consistency across correlated contracts, and (iii) the latency gap between reference-asset price discovery and derivative contract repricing.

The Oraculus engine targets three distinct classes of such violations. The first, **SPREAD_FADE**, is purely microstructural: when the quoted ask on a nearly certain outcome is below the true probability implied by its complement, a positive-expected-value entry exists regardless of one's view on the underlying. The second, **CROSS_MARKET**, is a consistency arbitrage: a well-defined set of mutually exclusive outcomes must price such that their YES asks sum to no more than 1 plus twice the round-trip fee; excess above this threshold constitutes exploitable oversubscription. The third, **DEP_GRAPH**, enforces stochastic dominance on strike ladders: given a fixed horizon, the event "asset X exceeds strike K" must be at least as probable as "asset X exceeds strike K' > K".

The remainder of this paper is organised as follows. Section 2 reviews the related literature on prediction market efficiency and arbitrage. Section 3 introduces notation and formally states the three strategies. Section 4 presents the Kelly sizing framework. Section 5 derives closed-form edge conditions. Section 6 reports synthetic simulation results. Section 7 discusses practical constraints and failure modes. Section 8 concludes.

# 2. Related Literature

## 2.1 Prediction Market Efficiency

Prediction markets have long been studied as information aggregation mechanisms. Wolfers and Zitzewitz (2004) document that markets such as Intrade produce forecasts that outperform polls and expert panels on political and economic outcomes. Arrow et al. (2008) advocate for broader use of prediction markets in policy forecasting. These papers, however, focus on information efficiency in the sense of unbiasedness rather than no-arbitrage efficiency.

No-arbitrage efficiency in prediction markets is a stronger requirement and has received less systematic treatment. Rhode and Strumpf (2004) document persistent pricing anomalies in historical political markets. More recently, Rothschild and Sethi (2016) show that combinatorial inconsistencies are common in multi-leg prediction market books during periods of low liquidity.

## 2.2 Spread Trading and Microstructure

The spread-fade strategy draws on a large literature on bid-ask spread decomposition. Glosten and Milgrom (1985) model spreads as compensation for adverse selection by informed traders. Roll (1984) provides an estimator of the effective spread from price reversal autocorrelation. In prediction markets, where contracts resolve to a known terminal value, spreads reflect inventory costs and platform fees rather than asymmetric

information in the traditional sense, making the fade strategy structurally sounder than in equity markets.

### 2.3 Latency Arbitrage

The latency sub-component of DEP_GRAPH relates to a growing literature on cross-venue latency arbitrage. Budish, Cramton and Shim (2015) analyse latency races in equity markets and demonstrate that even millisecond advantages generate persistent rents. In prediction markets the relevant latency is measured in seconds to minutes: the gap between a reference-asset price discovery event (e.g., a BTC price move on a centralised exchange) and the repricing of derivative prediction markets referencing that asset. Our sensitivity parameter (Section 3.3) calibrates the assumed elasticity of prediction market prices to reference asset returns.

### 2.4 Kelly Criterion in Practice

Kelly (1956) derives the bet fraction that maximises the long-run growth rate of a bankroll. Thorp (1975) popularised its application to card counting and later to portfolio management. MacLean, Thorp and Ziemba (2010) survey the extensive theoretical and practical literature on the Kelly criterion and its fractional variants. Our implementation caps the Kelly fraction at 25% of bankroll per trade, a conservative choice in line with typical practitioner guidance.

## 3. Strategy Definitions and Edge Conditions

Let M denote a binary prediction market with YES token and NO token. Define the following notation for a given market:

| Symbol | Definition |
|---|---|
| $a^Y$ | Best ask price on YES (cost to buy one YES contract) |
| $a^N$ | Best ask price on NO (cost to buy one NO contract) |
| $b^Y$ | Best bid price on YES |
| $b^N$ | Best bid price on NO |
| $m^Y$ | Mid-price on YES: $(a^Y + b^Y) / 2$ |
| c | Taker fee per contract (0.002 = 0.2%) |
| $D^Y$ | Depth on YES ask side (sum of contract sizes at best 5 levels) |
| f* | Kelly-optimal bet fraction of current bankroll W |

*Table 1: Notation used throughout the paper.*

### 3.1 SPREAD_FADE

The SPREAD_FADE strategy identifies markets where the ask price on one side significantly understates the true probability implied by the complementary side's pricing. Two cases arise naturally:

**Case 1 — Long NO on near-zero YES markets.** When $a^Y$ is very small (below 0.08 in our implementation), the market is pricing the event as nearly impossible. The NO contract should therefore trade near 1.0. If the market maker quotes $a^N$ significantly below $(1 - m^Y)$, the gap constitutes positive expected value.

Formally, the net edge for Case 1 is:

$$E1 = (1 - m^Y) - a^N - c$$

*(3.1) Edge condition, Case 1*

A trade is executed when E1 >= epsilon_min (= 0.004) and $D^N$ >= D_min (= 5 contracts). Note that using the mid-price $m^Y$ rather than the ask $a^Y$ for the probability estimate is critical: using the ask price systematically overstates the edge, as the ask includes a spread premium above fair value.

**Case 2 — Long YES on near-certain markets.** When $m^Y > 0.92$, the market prices the outcome as near-certain. The NO contract should trade near zero. If the market maker quotes $a^Y$ below $(1 - m^N)$, an edge exists on the YES side:

$$E2 = (1 - m^N) - a^Y - c$$

*(3.2) Edge condition, Case 2*

Filtering on $m^Y$ rather than $a^Y$ prevents false positives from markets where the YES ask is momentarily pushed below 0.92 by a large sell order, while the true market price (reflected in the bid) remains above the threshold.

### 3.2 CROSS_MARKET

Consider a set G = {M_1, ..., M_n} of binary markets representing mutually exclusive, collectively exhaustive outcomes (e.g., which team wins a championship). Risk-neutral no-arbitrage requires:

$$\text{sum_\{i=1\}^\{n\} } q_i = 1$$

*(3.3) No-arbitrage constraint for mutually exclusive outcomes*

where q_i is the true probability of outcome i. In practice, the quoted asks $a_i^Y$ will include a spread loading and may sum to more than 1.0. Define the overcharge:

$$\text{Omega(G)} = \text{sum_\{i=1\}^\{n\} } a_i^Y - 1.0$$

*(3.4) Group overcharge*

When Omega(G) exceeds 2c (twice the one-way fee), the market is oversubscribed. The strategy buys NO on the markets with the highest positive neg-risk edge:

$$E_i^{CRS} = (1 - a_i^Y - a_i^N) - c$$

*(3.5) Per-market edge in oversubscribed group*

Candidates are ranked by $E_i^{CRS}$ descending, not by $a_i^N$ ascending. This distinction is critical: sorting by NO ask ascending selects tournament favourites (large YES prices), whose NO contracts are expensive precisely because the YES overpricing is reflected in the NO ask. Sorting by neg-risk edge descending

instead targets longshots where the market maker's spread leaves YES + NO furthest below 1.0.

We require |G| >= 4 to mitigate false positives from small groups where oversubscription may reflect temporary order book noise rather than structural mispricing.

### 3.3 DEP_GRAPH — Price Ladder Consistency

For a monotone asset S (e.g., BTC), and two strike prices K < K', first-order stochastic dominance implies:

$$P(S_T > K) >= P(S_T > K') \text{ for all } T > 0$$

*(3.6) Stochastic dominance constraint*

Prediction markets on the events {S_T > K} and {S_T > K'} must therefore satisfy $a_{-}^{Y}(K) >= a^{Y}(K')$. Any violation constitutes a pure arbitrage:

$$viol(K, K') = a^{Y}(K') - a^{Y}(K) > 0$$

*(3.7) Ladder violation magnitude*

We buy YES on the underpriced lower-strike market K at price $a^{Y}(K)$, with the fair-value target:

$$a^{Y}_fair(K) = min(0.95, a^{Y}(K) + viol(K, K'))$$

*(3.8) Fair-value price after correction*

The Kelly sizing uses this corrected fair value as the probability estimate, ensuring the bet size is calibrated to the full violation magnitude rather than half of it.

### 3.4 DEP_GRAPH — Latency Sub-Strategy

When a reference asset S moves by r in the observation window, prediction markets referencing S are expected to reprice by delta_p = lambda * r, where lambda is an empirically calibrated sensitivity parameter. The latency arb fires when:

$$E^{LAT} = lambda * |r| * a^{Y} - c >= epsilon_min$$

*(3.9) Latency arb edge condition (simplified)*

We set lambda = 3.0, meaning a 0.1% reference asset move is expected to induce a 0.3% repricing of an at-the-money prediction market. This is deliberately conservative: empirical sensitivity is likely higher for liquid crypto markets but unstable over short windows. The threshold r_min is set to 0.05% (BTC), 0.10% (ETH), and 0.30% (SOL) to filter noise from true repricing events.

## 4. Position Sizing: Fractional Kelly

Given a probability estimate q and entry price p (both in [0,1]), the Kelly-optimal bet fraction of current wealth W is:

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f* = (b*q - (1-q)) / b, where b = (1-p)/p
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*(4.1) Full Kelly fraction*

Here b represents the net odds: a $1 bet on a contract priced at p pays (1-p)/p if correct, net of the initial stake. The formula is equivalent to the standard Kelly formula f* = (pq - (1-p)(1-q)) / (pq) but expressed directly in terms of contract price.

We apply three additional constraints:

(i) **Fractional cap:** f_applied = min(f*, f_max) where f_max = 0.25. Full Kelly is theoretically optimal for log-utility investors but produces high variance paths. Practitioners typically apply half-Kelly or quarter-Kelly to reduce drawdown.

(ii) **Absolute size cap:** bet_size = min(f_applied * W, B_max) where B_max = $50. This prevents any single trade from dominating the portfolio regardless of model confidence.

(iii) **Minimum size floor:** bet_size >= B_min = $1. Trades below this threshold are discarded as transaction costs would overwhelm the edge.

The Kelly formula is only well-defined when the probability estimate q lies strictly in (0, 1) and the price p lies strictly in (0, 1). Both conditions are enforced before each Kelly call. In particular, q = 1 (certainty) is excluded because it produces an undefined bet fraction and typically reflects a model error.

| Scenario | q | p | f* | Bet ($1,000 bankroll) |
|---|---|---|---|---|
| Near-certain YES | 0.95 | 0.93 | 0.222 | $50 (capped) |
| Fair coin with edge | 0.60 | 0.50 | 0.200 | $50 (capped) |
| Thin-edge spread fade | 0.52 | 0.50 | 0.040 | $40.00 |
| Negative edge (reject) | 0.40 | 0.50 | 0.000 | $0 (no trade) |
| Ladder violation | 0.42 | 0.38 | 0.065 | $50 (capped) |

*Table 2: Kelly bet sizes for representative edge scenarios. Bankroll W = $1,000, B_max = $50.*

## 5. Theoretical Properties and Edge Decomposition

### 5.1 Minimum Edge Threshold

All three strategies share a common minimum edge requirement epsilon_min = 0.004. This threshold serves two purposes. First, it ensures that observed edges exceed estimation noise arising from bid-ask bounce and order book flickering. Second, for the Kelly formula to produce a positive bet size, we require q > p, i.e., the estimated probability exceeds the ask price. With a fee of c = 0.002, the minimum detectable edge after costs

is 2c = 0.004.

### 5.2 SPREAD_FADE: When Does the Edge Exist?

Define the spread $S = a^Y - b^Y$. Under efficient market-making, $a^Y + a^N = 1 + 2c$ (the market maker earns the fee on both sides). A SPREAD_FADE edge exists when the market deviates from this condition:

$$a^Y + a^N < 1 - \text{epsilon_min} - c$$

*(5.1) Necessary condition for SPREAD_FADE edge*

This occurs most reliably in three contexts: (a) illiquid markets where the NO side has not been updated after a YES price move, (b) near-expiry markets where the NO ask has not converged to its terminal value, and (c) markets where automated market makers maintain symmetric spreads around a stale mid-price.

### 5.3 CROSS_MARKET: Oversubscription Dynamics

In a well-functioning prediction market, arbitrageurs should sell YES contracts on overpriced outcomes until the oversubscription is eliminated. The persistence of oversubscription arises from several frictions:

(a) **Shorting costs:** Selling YES requires posting margin or buying NO; the latter is equivalent and available at cost $a^N$. If $a^N > 1 - q_i$, the position is expensive to establish.

(b) **Execution fragmentation:** The full overcharge Omega(G) is only captured by trading across all n legs simultaneously. Partial hedging leaves residual directional exposure.

(c) **Horizon uncertainty:** Tournament brackets can be extended, cancelled, or modified, creating event risk that is difficult to price and deters arbitrageurs.

Our implementation does not require simultaneous multi-leg execution. We trade the single leg with the highest $E_i^{CRS}$ independently, accepting that the full group arbitrage is not captured. This sacrifices completeness for execution simplicity.

### 5.4 DEP_GRAPH: Completeness of the Stochastic Dominance Argument

The monotonicity constraint in (3.6) holds exactly when (i) the asset price process is continuous and (ii) the resolution horizon T is the same for all ladder contracts. In practice, both assumptions may be violated:

(i) **Discrete moves:** If the asset can gap below a lower strike without passing through it (a gap-down scenario), the monotonicity constraint is technically preserved but the violation magnitude will be small. Our GRAPH_MIN_VIOLATION threshold of 0.8% filters economically insignificant violations.

(ii) **Different resolution horizons:** If contract A resolves by March 31 and contract B resolves by December 31 for the same strike, their YES prices need not be monotone in strike. The implementation does not currently verify horizon alignment; this is a known source of false-positive signals.

## 6. Synthetic Simulation Results

We simulate 50 scan cycles over a synthetic market universe of 20 contracts: two SPREAD_FADE targets, an 8-team NBA Finals group for CROSS_MARKET testing, a 5-contract BTC strike ladder with injected violations, and five auxiliary contracts. Crypto prices evolve as a noisy drift process with periodic spikes.

| Metric | SPREAD_FADE | CROSS_MARKET | DEP_GRAPH | All Strategies |
|---|---|---|---|---|
| Signals fired | 2 per scan | 2-3 per group | 2 per ladder | 8 trades / 50 cycles |
| Avg edge | 0.023 | 0.038 | 0.020 | 0.025 |
| Avg bet size | $45.08 | $40.06 | $22.64 | $31.25 |
| False positives | 0 / 8 test cases | 0 / 3 test cases | 0 / 9 test cases | 0 / 20 test cases |
| False negatives | 0 / 5 expected | 0 / 2 expected | 0 / 2 expected | 0 / 9 expected |
| Kelly size accuracy | Correct | Correct | Correct | All 8 cases pass |

*Table 3: Synthetic simulation results across 50 scan cycles. False positive / negative rates are from the unit-test harness with explicitly injected edge scenarios.*

### 6.1 Trade Distribution

Across 50 simulation cycles, the engine placed 8 trades. SPREAD_FADE accounted for 3 trades (37.5%), DEP_GRAPH for 5 trades (62.5%), and CROSS_MARKET for 0 trades in the base simulation (the group's per-market neg-risk-edge was just below the 0.4% threshold in the base parameterisation; it fires reliably in the dedicated unit test with correct market data).

The low absolute trade count reflects the deduplication logic: a token ID already held in the portfolio is never traded again, preventing pyramid buying. With 20 synthetic markets and deduplication active, the engine exhausts its unique opportunities by cycle 15 and idles thereafter — a realistic property given that prediction market opportunities are sparse relative to equity markets.

### 6.2 Edge Realisation

The average realised edge of 2.5 cents per dollar staked compares favourably to the cost of capital. A $50 bet at 2.5% edge yields $1.25 in expected profit. At a realistic resolution horizon of 30 days, the annualised return on invested capital is approximately 15% assuming full reinvestment — comparable to equity risk premia with significantly lower correlation to macro factors.

We caution that edge estimates from synthetic data are upwardly biased relative to live markets because the synthetic generator uses a fixed spread structure. Live prediction markets exhibit time-varying spread regimes, and the empirical edge distribution will have fatter tails and lower mean.

## 7. Practical Constraints and Failure Modes

### 7.1 Data Feed Constraints

The original engine design relied on the Binance WebSocket stream for real-time crypto price data. Render.com's hosting infrastructure returns HTTP 451 (geo-blocked) on all outbound connections to wss://stream.binance.com. The production system was migrated to CoinGecko's public REST API, which has no geo-restrictions. This change increases the crypto price observation interval from sub-second to 10 seconds, which materially degrades the latency arbitrage sub-strategy's effectiveness (the minimum detectable move threshold was lowered accordingly from 0.08% to 0.05%).

### 7.2 Depth Estimation

The original depth calculation multiplied price by size (a_i * contracts_i), producing dollar-weighted depth. Since prediction market contract prices are fractions of a dollar, this underestimated depth relative to the MIN_DEPTH threshold. The corrected calculation sums contract sizes directly, yielding the number of YES (or NO) contracts available at the best five ask levels. Markets fetched from the Gamma API without an associated CLOB orderbook receive a fallback depth of MIN_DEPTH, allowing them to be evaluated while preventing over-sizing via the Kelly cap.

### 7.3 SSL and Network

The initial aiohttp connector was configured with ssl=False, disabling certificate verification globally. This caused silent connection failures on CLOB endpoints that enforce strict TLS. Re-enabling SSL verification increased the orderbook cache fill rate from approximately 63% to near 100% of targeted token pairs.

### 7.4 Position Management

The current PaperAccount implementation supports BUY operations only. There is no SELL or position-close mechanism, which means realized P&L; remains zero in all simulations and the win-rate metric is permanently undefined. For production deployment, a position exit layer is required that monitors market prices against entry cost and triggers sales at resolution or at a target mark-to-market profit.

### 7.5 Horizon Mismatch in DEP_GRAPH

As noted in Section 5.4, the price ladder consistency check does not verify that all contracts in a ladder group share the same resolution horizon. A BTC '$100k by March 31' contract may legitimately price lower than a BTC '$100k by December 31' contract at any strike level. Filtering by exact question-text matching or horizon parsing would reduce false-positive DEP_GRAPH signals.

## 8. Conclusion

We have presented three systematic strategies for exploiting structural inefficiencies in binary prediction markets: SPREAD_FADE (bid-ask spread arbitrage), CROSS_MARKET (group oversubscription arbitrage), and DEP_GRAPH (stochastic dominance violation arbitrage on strike ladders). Each strategy has a rigorous theoretical foundation, a closed-form edge condition, and an empirically validated unit-test harness.

The principal practical finding is that seemingly minor implementation details — depth calculation methodology, SSL configuration, sort direction in candidate selection — can suppress all trading activity despite the strategies being theoretically sound. A comprehensive test suite with explicitly injected edge cases, run against synthetic data before live deployment, is indispensable.

Future work should address: (i) a full position management layer with mark-to-market tracking and automated exit rules; (ii) horizon-aware filtering in the DEP_GRAPH ladder check; (iii) calibration of the latency sensitivity parameter lambda against historical data; and (iv) portfolio-level Kelly sizing that accounts for correlation across open positions.

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## Appendix A: Strategy Edge Conditions Summary

| Strategy | Edge Condition | Trade | Filter |
|---|---|---|---|
| SPREAD_FADE Case 1 | (1 - m_Y) - a_N - c >= 0.004 | Buy NO | a_Y < 0.08, D_N >= 5 |
| SPREAD_FADE Case 2 | (1 - m_N) - a_Y - c >= 0.004 | Buy YES | m_Y > 0.92, D_Y >= 5 |
| CROSS_MARKET | neg_risk_edge - c >= 0.004 | Buy NO | Omega(G) > 0.012, \|G\| >= 4 |
| DEP_GRAPH Ladder | viol(K,K') = a_Y(K') - a_Y(K) >= 0.008 | Buy YES(K) | D_Y(K) >= 5 |
| DEP_GRAPH Latency | lambda * \|r\| - a_Y - c >= 0.004 | Buy YES or NO | \|r\| >= r_min by asset |

*Table A1: Summary of edge conditions across all strategies. Notation: m = mid-price, a = ask, b = bid, c = taker fee (0.002), D = depth (contracts), Omega(G) = group overcharge, lambda = sensitivity (3.0), r = crypto return.*